Name : Ahmed Adel Elsayed
ID : 192300594

Motor Driver Selection

1. Introduction

In the smart wheelchair project, a motor driver is required to control the DC motors that move the wheels. The Arduino microcontroller cannot drive motors directly because its output pins provide only a small current. Therefore, a motor driver is used as an interface between the Arduino and the motors.

The motor driver receives low-power control signals from the Arduino and supplies high current and voltage to the motors from the battery.

2. System Power and Motor Requirements

Battery: 24V battery

Motors: DC motors used to drive the wheelchair wheels

Controller: Arduino microcontroller

Because the motors require relatively high current to move the wheelchair load, a high-current motor driver must be selected.

3. Motor Driver Options Considered

1. BTS7960 Motor Driver

2. L298N Motor Driver

3. Sabertooth 2×32 Motor Driver

4. Comparison Between Motor Drivers

Feature	BTS7960	L298N	Sabertooth 2×32
Maximum Current	43A peak	2A	32A per motor
Voltage Range	5.5V – 27V	5V – 35V	Up to 30V
Efficiency	High	Low	Very High
Motor Power Support	High power motors	Small motors	Professional robotics
Price in Egypt	125 – 520 EGP	90 – 150 EGP	5000 – 7000 EGP
Availability in Egypt	Available	Very available	Rare

5. Technical Overview of the Selected Driver (BTS7960)

Operating voltage: 5.5V – 27V

Continuous current: 27A

Peak current: 43A

PWM frequency: up to 25 kHz

Protection features: Over-current protection, Thermal shutdown, Undervoltage protection

The module uses two BTS7960 chips to create a full H-bridge circuit for bidirectional motor control.

6. Price and Availability in the Egyptian Market

BTS7960 Motor Driver: 125 – 520 EGP

L298N Motor Driver: 90 – 150 EGP

Sabertooth Driver: 5000+ EGP

7. Why BTS7960 Was Selected

1. High current capability (up to 43A peak).

2. Compatible with 24V battery systems.

3. Supports PWM control with Arduino.

4. Built-in safety protections.

5. Lower cost compared to professional drivers.

8. Conclusion

After evaluating several motor driver options, the BTS7960 motor driver was selected because it provides the required current capacity, supports the 24V battery system, and is affordable and available in the Egyptian market.

This driver allows efficient control of the wheelchair motors using Arduino and is a critical component in the smart wheelchair control system.